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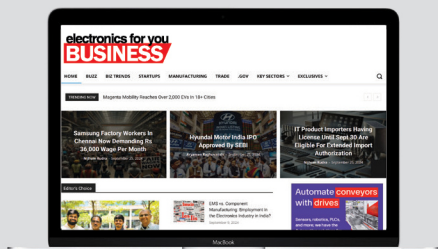


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Practical Thinking: Shark Fin antenna-Equipped with Diotec Voltage Regulator LDI51XXEN-AQ.

A shark fin antenna works as a multifunctional communication device receiving and transmitting signals for AM/FM radio, GPS, cellular networks, Wi-Fi and satellite services. It integrates multiple antennas into a compact housing ensuring minimal signal interference and optimized performance. Shark Fin antennas are designed to be weather-resistant, capable of operating in extreme temperatures from freezing conditions to scorching heat. This durability ensures that the antenna performance is not obstructed by environmental factors, maintaining reliable communication in all conditions.



A Low Dropout Regulator (LDO) is an essential component in a shark fin antenna ensuring a stable and regulated voltage supply for various internal circuits like GPS, radio, cellular (4G/5G) and Wi-Fi modules. The shark fin antenna is powered by the vehicle 12V battery but many of its internal modules require lower stable voltages like 3.0V, 4.0V or even 1.8V. LDO (LDI51-XXEN-AQ) Converts 12V from the vehicle to suitable lower voltages.

The Diotec LDI51XXEN-AQ is a high-performance Low-Dropout (LDO) regulator designed for efficient power management in Automotive. The LDI51-4.0EN-AQ specifies a typical dropout voltage of 120mV when the output current is 500mA and the output voltage is 4.0V with integrated current limiting and thermal overload protection. The low quiescent current 1uA extend the life. Input voltage is up to 18V, Power dissipation is 500mW and junction temperature is -40 to +125°C.

FOR MORE INFO



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